

DATABASE MANAGEMENT SYSTEM (DBMS): MASTER TEXTBOOK

1. Introduction to DBMS

What is DBMS?

A DBMS is a software system designed to define, manipulate, retrieve, and manage data in a database. Its goal is to provide efficient, reliable, and secure data storage.

2. Entity-Relationship (ER) Model

Core

Entity: A real-world object (e.g., Student).
Attribute: Characteristics of an entity (e.g., Name).
Relationship: Associations between entities (e.g., enrolls).

Core Components

Entity: A real-world object (e.g., Student, Course).

Attributes: Characteristics of an entity (e.g., Name, ID).

Relationships: Associations between entities (e.g., Student *enrolls in* Course).

3. The Relational Model

Tables and Keys

Data is organized in tables (relations).

Primary Key: Uniquely identifies a record in a table.

Foreign Key: Establishes a link between two tables.

Core Commands

- **DDL:** CREATE, ALTER, DROP.
- **DML:** SELECT, INSERT, UPDATE, DELETE.
- **DCL:** GRANT, REVOKE.

4. Normalization

Goal: Reduce Redundancy

Normalization is the process of organizing data to reduce redundancy and improve data integrity (1NF, 2NF, 3NF, BCNF).